Template Solutions Document

Contents

Evaluated criteria

- C1. Example criterion description.
- C2. Example criterion description.
- C3. Example criterion description.

1 (5) Example Uniform Investment Plan (C1)

A variable V follows a uniform model on $[8, 16]$. Use the graph and density model to answer the tasks.

Questions/Tasks.

a) **(2)** Write the density function and support.

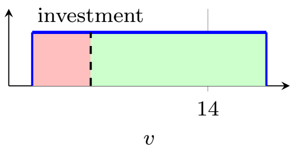
b) **(3)** Compute $P(10 \leq V \leq 14)$ and justify with area.

a) (2) Write the density function and support.

$$f_V(v) = \begin{cases} \dfrac{1}{16-8} = \dfrac{1}{8}, & 8 \leq v \leq 16, \\ 0, & \text{otherwise.} \end{cases}$$

$$8 \leq v \leq 16 \quad \longrightarrow \quad f_V(v) = \frac{1}{8} \quad \longrightarrow \quad \int_8^{16} f_V(v)\, dv = 1$$

b) (3) Compute $P(10 \leq V \leq 14)$ and justify with area.



$$P(10 \leq V \leq 14) = \int_{10}^{14} \frac{1}{8}\,dv$$
$$= \frac{1}{8}(14-10) = \frac{4}{8} = 0.5$$

The shaded area represents half of the support interval.

2 (6) Example Linear Density Model (C2)

Suppose X has density $f(x) = kx$ for $0 \leq x \leq 2$.

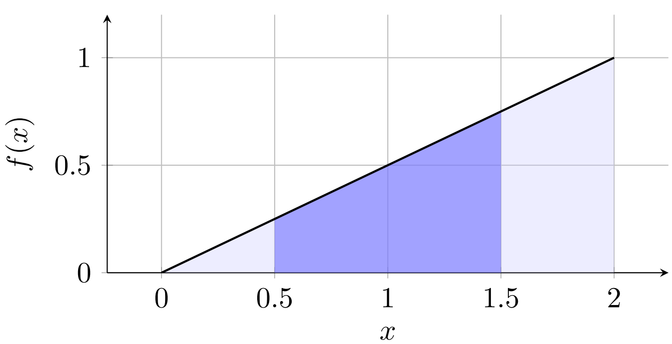
Questions/Tasks.

a) **(3)** Find the value of k.

b) **(3)** Compute $P(0.5 \leq X \leq 1.5)$.

a) (3) Find the value of k.

$$1 = \int_0^2 kx\,dx = k\left[\frac{x^2}{2}\right]_0^2 = 2k \qquad \Rightarrow \qquad k = \frac{1}{2}.$$



b) (3) Compute $P(0.5 \leq X \leq 1.5)$.

$$P(0.5 \leq X \leq 1.5) = \int_{0.5}^{1.5} \frac{1}{2}x\,dx = \left[\frac{x^2}{4}\right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{4} = 0.5$$

3 (4) Example Table and Probability Summary (C3)

A summary table is shown for two intervals of a piecewise density.

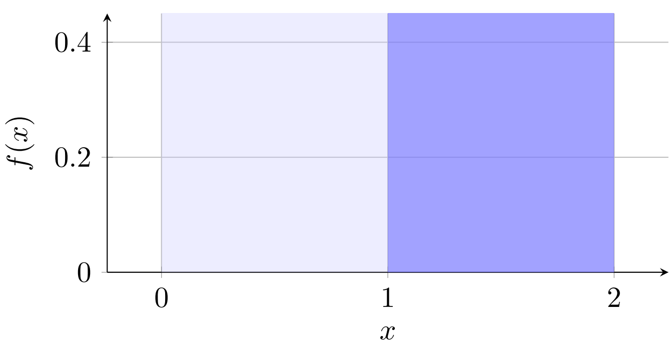
Questions/Tasks.

a) **(2)** Complete the probability summary table.

b) **(2)** State one consistency check.

a) (2) Complete the probability summary table.

Interval	Expression	Result
$[0,1]$	$\int_0^1 0.4\,dx$	0.4
$[1,2]$	$\int_1^2 0.6\,dx$	0.6
Total	$0.4+0.6$	1



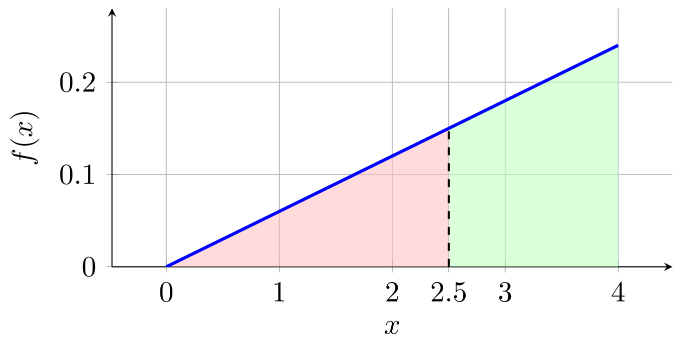
b) (2) State one consistency check.

A valid check is that all interval probabilities are nonnegative and add to 1.

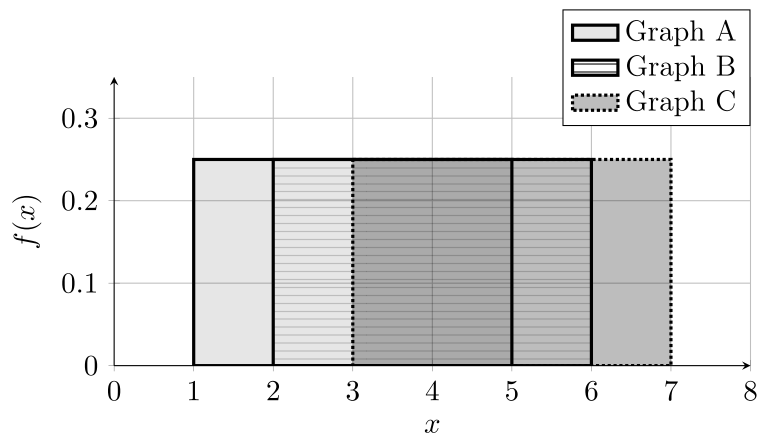
4 (0) Graph Command Examples

This section provides one representative command from each graph file in `book/preamble/graphs`.

a) (0) Example from `uniform_linear_probability_distributions.tex`.



b) (0) Example from `probability_distribution_graphs.tex`.



c) (0) Example from `normal_distribution_graphs.tex`.